

PHILIP LODONU

Frontend Engineer & AI Engineer

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• Accra, Ghana

SUMMARY

Software Engineer with hands-on experience building responsive, user-focused web and mobile applications using JavaScript, React.js, React Native, Node.js, and Tailwind CSS. Experienced in developing end-to-end applications, integrating AI capabilities, and using modern development tools to build practical software solutions. Completed software engineering training at Codetrain Africa and continues to strengthen skills through real-world projects and professional experience. A problem solver and collaborative team player with a strong interest in frontend development, AI engineering, and creating technology that solves real-world problems.

EXPERIENCE

Frontend Engineer — Aerolabgh(Volunteering)

Accra, Ghana

- Built a dynamic and interactive single-page application (SPA) using React
- Organized bi-weekly meetings with clients to communicate project progress and align on further deliverables
- Designed and styled the user interface using Tailwind CSS, creating a clean, modern look that works smoothly on all devices
- Developed page layouts and custom navigation using Tailwind CSS, DaisyUI and React Icons

Web Developer — Oasis Infobyte (Internship)10/2024 - 11/2024

Delhi, India

- Assisted in developing a responsive company website, improving mobile usability and user experience
- Engaged in weekly code reviews, strengthening skills in HTML, CSS, and JavaScript
- Took part in daily team meetings and work sessions, improving project delivery speed by 10%
- Communicated insights and recommendations effectively to the broader team and stakeholders through well-crafted presentations, enhancing stakeholder understanding by 60%. Tools used: PowerPoint, Canva
- Completed assigned tasks on time and ensured timely submission

Frontend Engineer — Complete Farmer (Internship)06/2024 - 08/2024

Airport, Accra

- Fixed bugs in backend email templates and the agent recruitment website, improving reliability, functionality, and cross-platform consistency
- Collaborated with the team to build modular, reusable UI components inspired by ShadCN, promoting scalability and design consistency
- Assisted in resolving frontend issues related to responsiveness, styling, and component integration using React.js, Tailwind CSS, and TypeScript
- Collaborated with product managers to test new apps and websites during the onboarding stage, documenting and reporting issues before deployment

- Participated in daily sprints, weekly syncs, and technical workshops, contributing to smooth project planning and team alignment
- Completed assigned tasks within deadlines, ensuring timely submission
- Communicated effectively with teammates and stakeholders, contributing to improved teamwork and decision-making

SKILLS

Frontend Development: HTML, Vanilla CSS, Tailwind CSS, NativeWind, React.js, React Native, Next.js, Chakra UI, Vue, Zustand, TanStack React Query

Programming Languages: JavaScript, TypeScript, Python

Databases & Storage: MongoDB Atlas, Firebase, Redis, ChromaDB, Cloudinary

Backend Development: Node.js, Express.js, Firebase Authentication, RESTful APIs, Server-Sent Events (SSE), Socket.io, Nodemailer, FastAPI, APIs

Version Control: Git, GitHub

Development Tools & Platforms: VS Code, Postman, Claude, Antigravity, Claude Code, Render, Vercel, Paystack

Soft Skills: Problem-solving, Team Collaboration, Communication, Adaptability

AI & Data Technologies: LLMs, AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search

PROJECTS

Nursify

- Built a comprehensive digital health platform connecting patients with verified nursing professionals for appointments, telehealth consultations, and continuous health monitoring
- Developed role-based dashboards for Patients, Nurses, and Admins, ensuring secure access control with tailored interfaces and permissions for each user type
- Implemented real-time communication using Socket.io, enabling instant messaging and video call consultations between patients and nurses
- Designed interactive health data visualizations to track patient vital signs over time, providing clear insights into health trends
- Created specialized Maternal Health and Child Health modules with structured UI flows for managing prenatal care, immunizations, and child health records
- Built a secure document management system allowing users to upload, store, and view medical documents such as prescriptions and lab results
- Developed a full appointment booking system enabling patients to search for nurses, schedule sessions, manage appointments, and submit reviews
- Designed and implemented a scalable RESTful API using Node.js, Express.js, and MongoDB to manage users, appointments, health records, and notifications

- Implemented a nurse verification and credentialing system, including license validation and profile management to ensure platform trust and safety
- Implemented automated background tasks and notifications for appointment reminders, system updates, and user alerts
- Built a secure appointment booking system enabling nurse discovery, scheduling, and patient reviews
- Integrated Capacitor to package and deploy the web application as a native Android mobile application, enabling access to device-level capabilities and mobile deployment.

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SkyPay

- Built SkyPay, a full-stack digital wallet application enabling secure deposits, withdrawals, and peer-to-peer transfers with a focus on the African mobile money ecosystem
- Architected a scalable React Native (Expo) mobile application using file-based routing and a clean separation of UI, state, and service layers
- Developed a high-performance financial dashboard with real-time balance updates, privacy mode (balance masking), and animated, time-based greetings
- Implemented efficient state and server data management using Zustand and React Query (TanStack Query) for caching, synchronization, and optimistic updates
- Designed a modern, premium UI using NativeWind (Tailwind CSS), incorporating gradients, glassmorphism effects, and smooth animations
- Designed and implemented a robust financial transaction engine using Node.js, Express.js, and MongoDB with ACID-compliant transactions to prevent double-spending and data inconsistencies
- Integrated Mobile Money providers (MTN, Vodafone, Tigo) with strict validation for phone numbers, networks, and transaction rules
- Implemented a secure double-entry transaction ledger system, ensuring all balance changes are traceable, immutable, and auditable
- Applied strong security practices using JWT-based authentication, bcrypt password hashing, and role-based access control middleware

Kente AI — Ghana Culture & Tourism AI Assistant

- Built Kente AI, a web app that acts like a smart Ghana travel guide: it plans trips, suggests places to visit based on your interests, and teaches basic phrases in Twi, Ga, and Ewe
- Made AI replies show up gradually as they're generated, instead of waiting for the whole answer, to improve the user experience and make the app feel faster and more responsive
- Added the ability to upload a photo of a Kente cloth pattern, a landmark, or a Ghanaian dish and have the AI recognize what it is and explain its cultural background
- Built a data pipeline that cleans up and organizes raw information so the AI can search through it accurately

- Set up a system that lets the AI search through that information by meaning, not just keywords, so it finds the most relevant answers
- Built a Q&A system that forces the AI to only answer using verified information it retrieved, instead of making things up
- Made sure the AI handles Ghana's different regions, ethnic groups, and languages respectfully, and taught it to say "I'm not sure" instead of guessing

E-Commerce Platform

- Architected a full-stack luxury fashion and beauty e-commerce platform using React 19, Vite, Express.js, and MongoDB, decoupling the customer storefront from a dedicated administrative management dashboard.
- Integrated the Paystack API for online card and Ghanaian Mobile Money (MTN, Telecel, AirtelTigo) payments, engineering webhook listeners with HMAC signature verification and payment verification fallbacks.
- Engineered an interactive delivery location pin-pointing system using Leaflet.js, CartoDB Dark Matter map tiles, and the HTML5 Geolocation API to attach precise GPS coordinates and Google Maps links to customer orders.
- Implemented frontend state management and data fetching using Zustand and TanStack React Query v5, supporting persistent shopping carts, wishlists, infinite-scroll catalog navigation, dynamic sorting, subcategory filtering, and client-side caching.
- Developed an interactive “Beauty & Style Match” recommendation quiz that evaluates skincare, makeup, and fragrance preferences to dynamically generate personalized product recommendations.
- Designed and implemented an administrative analytics dashboard using Recharts to track revenue trends, category performance, top-performing products, inventory levels, and customer metrics.
- Engineered a comprehensive product and inventory management system supporting multi-image Cloudinary uploads via Multer, automated SKU generation, stock adjustments, and product variants across sizes and colors.
- Implemented customer relationship management and order fulfillment functionality featuring order lifecycle tracking, customer tagging, administrative notes, account management, and bulk CSV exports for orders, invoices, and transactions.
- Engineered a transactional communication and document system using Nodemailer and EJS templates to deliver order receipts, shipping updates, back-in-stock alerts, and automated invoices.
- Developed an administrative promotion and customer loyalty framework supporting promo code validation, discount calculations, gift card redemptions, return requests, and tier-based loyalty point accrual.
- Architected a role-based access control (RBAC) system using Firebase Authentication, JWTs, and Bcrypt.js, securing administrative operations including store settings, admin user management, maintenance controls, and audit logs.
- Implemented real-time administrative notifications and backend reliability mechanisms using Socket.io, Helmet, Express Rate Limit, and Node-Cron background tasks.

NutriGhana

- Developed a responsive nutrition and local food application using React Native and React.js, optimized for Android devices with a focus on performance and accessibility

- Implemented secure authentication and backend integration using Firebase, Express.js, and MongoDB, supporting user account management and personalized experiences
- Built reusable interfaces for displaying local food information, nutritional data, and personalized health tips across mobile and web environments
- Designed a consistent cross-platform UI using Tailwind CSS and NativeWind, incorporating responsive layouts, reusable components, and smooth animations

Todo-List

- Developed a responsive and reusable web interface using React.js and Tailwind CSS, focusing on clean component architecture, responsive design, and user-friendly interactions

Tic-Tac-Toe Game

- Developed and deployed a responsive two-player web game using HTML, CSS, and JavaScript, implementing interactive gameplay and dynamic DOM-based updates

EDUCATION

Codetrain Africa	Accra, Ghana
Software Engineering	Apr 2023 – Present
Keta Senior High School	Keta, Ghana
Senior High School Education	Sep 2022 – Nov 2024

Certifications

Simplilearn	Introduction to MERN STACK
SoloLearn	Introduction to Javascript

LANGUAGES

English Proficient ●●●●○ **Ewe** Native ●●●●● **Akan** Native ●●●●●